EBC 460 CHANA CD101

TRW End of Line Proposal
TRW下线检测建议

Revision 0.5

Date: October 08th 2011

by O. Brehm

© TRW Automotive Holdings Corp. 2011

Revision Log.

Version	Date	Name	Section affected	Changes
0.1	2011.07.28	Oliver Brehm		Initial Version
0.2	2011.08.17	O. Brehm	All	Add Chinese translation
0.3	2011.08.24	O. Brehm	Procedure proposal	Substitute low speed test with high speed test
0.4	2011.10. 24	O. Brehm	Diagnostic Communication	changed Routine Control
0.5	2011. 11.08	O. Brehm	Sensor Calibration	Added section Sensor Calibration

DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

Sensor Calibration (for ESC, only)

To bring the ESC – system to fault-free condition, it is necessary to calibrate the IIS-sensor and the Steering angle sensor. This process has to be performed before running the Roller bench Test.

The preferred position during the assembly process in manufacturer plant is Wheel Alignment Rig (especially for the Steering Angle Sensor) or the area in front of Roller bench Test.

Condition for calibration:

Vehicle speed < 3kmph

No Control Algorithm is Active

At least one Wheel Speed Sensor is valid

Vehicle is in standstill on a flat surface and no fault is stored, except Fault-Id :

180 ($50 $12 $08)	Steering Angle Sensor offset calibration missing.
190 ($53 $63 $54)	Yaw sensor offset calibration missing.
202 ($53 $61 $54)	Lateral sensor offset calibration missing.
264 ($53 $64 $54)	Longitudinal sensor offset calibration missing.

DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

Steering Angle Sensor Calibration (for ESC, only)

Diagnostic command (available during EOL session $10 $04):

$31 $01 $30 $27 Steering Angel Sensor Offset determination

$71 $01 $30 $27 positive response

Result request of Steering Angle Sensor Offset determination

$31 $03 $30 $27 Result request of IIS Sensor Offset determination

$71 $03 $30 $27 $XX

$XX – $00 – Routine successful

$01 – Routine running

$02 – Routine failed

DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

IIS - Calibration (for ESC, only)

Diagnostic command (available during EOL session $10 $04):

$31 $01 $30 $21 IIS - Offset determination

$71 $01 $30 $21 positive response

Result request of IIS Offset determination

$31 $03 $30 $21 Result request of IIS Sensor Offset determination

$71 $03 $30 $21 $XX $XX – $00 – Routine successful

$01 – Routine running

$02 $YY $ZZ – Routine failed, where $YY gives the sensor ID and $ZZ gives the failure type

$YY – Sensor decoding

$01 – Yawrate sensor

$02 – Lateral sensor

$03 – Longitudinal sensor

$ZZ – failure type decoding

$01 – Signal not valid

$02 – Sensor signal out of range

$03 – Internal fault

DRAFT

In case of - Routine failed -, bring vehicle to repair and fix the fault!

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

Purpose/目的

The dynamic portion of vehicle test is seeking to detect the following installation faults:

- Crossed brake line plumbing.
- Crossed wheel speed sensor wiring.

整车测试的动态部分是为了检测以下安装错误：

- 错误的制动管路连接
- 错误的轮速传感器线束连接

In order to perform this detection, each wheel speed is altered independent of the other wheels, and the module speeds are monitored to verify that the speeds change in the expected order.

为了进行这些检查，各个轮速独立于其它轮速进行变化，模块会监控轮速变化是否按照期望的顺序进行。

In case that a low speed test is performed the brake torque which is measured by the roller bench will be used as an indicator.

在进行低速测试的情况下，转鼓测量的制动力矩将作为一个指示值。

DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

Test Step 测试步骤 / Test Scope测试范围:

- **Step1/第一步:** Communication Set Up/ 建立通讯连接
- **Step2/第二步:** ECU – Identification/ ECU识别
- **Step3/第三步:** Read DTC---Clear DTC---Read DTC/ 读错误码—清错误码—读错误码
- **Step4/第四步:** Check process byte / 检查过程字节
- **Step5/第五步:** Check input / output signal/ 检查输入输出信号
- **Step6/第六步:** Wheel Speed Signal test/ 轮速信号测试
- **Step7/第七步:** Brake and ABS test/ 制动器和ABS测试
  High Speed Dynamic test preferred / 高速动态测试（优选）
  Low Speed Dynamic test alternatively / 低速动态测试（可选）
- **Step8/第八步:** Read DTC---Clear DTC---Read DTC/读错误码—清错误码—读错误码
- **Step9第九步:** Warning Light test/ End of test/ 仪表警告灯测试及结束测试

DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

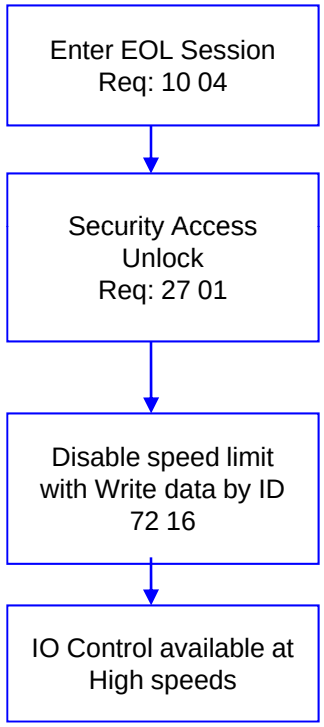
Diagnostic Communication诊断通讯/ Valve activation request电磁阀激励请求

Diagnosis via CAN/ 通过CAN进行诊断

Conditions for I/O control/ 输入输出控制的条件

- Only in EOL vehicle manufacturer session (for v < 3kph; entry condition)
  仅对主机厂的下线检测部分（车速小于3Kph，进入条件）
- No active control algorithm (ABS / Degas pulse…)
  电磁阀不被激励
- Remember to activate Tester Present to keep module in EOL session
  激发诊断设备，保持模块处于EOL模式
- I/O command expires after 4seconds, can be retriggered
  输入输出命令4秒钟后过期，可以被重新触发
- I/O control will be inactive if vehicle speed exceeds v = 10kph
  如果车速超过10Kph，输入输出控制将会关闭
- No under- or over-voltage (12V – 13,5V; 10,5A)
  电压不能过低或过高(12V – 13,5V; 10,5A)
- No valve errors or wheel speed-sensor fault present
  不能有阀或者轮速传感器的错误存在

I/O Control at high speeds



To allow I/O control on higher speeds the speed limit has to be disabled: $2E $72 $16 $AA

如果需要在更高速度下进行输入输出控制，速度限制应被关闭：$2E $72 $16 $AA

DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

Diagnostic Communication诊断通讯/ Valve activation request电磁阀激励请求

SID: $31 - Routine Control/ 诊断服务$31-例程控制

ID No	ID Name	Routine Control Type	Byte	Bit	Signal/ Variable name	Bit Description	0	1
30 28	EOL test by Routine Control	01: Start Routine 02: Stop Routine 03: Request Results	1	7	Valve activation current state	RR iso	off	on
				6		RR dump	off	on
				5		LR iso	off	on
				4		LR dump	off	on
				3		RF iso	off	on
				2		RF dump	off	on
				1		LF iso	off	on
				0		LF dump	off	on
			2	5-7	Valve activation current state	reserved		
				4		Pump motor	off	on
				3		RFLR TC iso (VSC only) 0 for ABS	off	on
				2		LFRR TC iso (VSC only) 0 for ABS	off	on
				1		RFLR TC supply (VSC only) 0 for ABS	off	on
				0		LFRR TC supply (VSC only) 0 for ABS	off	on
			3		Waiting time (high byte)	Range: 0x000 – 0x01FF 0ms – 5110 ms Steps: 1bit = 10ms Resolution: 10 ms		
			4		Waiting time (low byte)			
			5	7	Valve activation next state	RR iso	off	on
				6		RR dump	off	on
				5		LR iso	off	on
				4		LR dump	off	on
				3		RF iso	off	on
				2		RF dump	off	on
				1		LF iso	off	on
				0		LF dump	off	on
			6	5-7	Valve activation next state	reserved		
				4		Pump motor	off	on
				3		RFLR TC iso (VSC only) 0 for ABS	off	on
				2		LFRR TC iso (VSC only) 0 for ABS	off	on
				1		RFLR TC supply (VSC only) 0 for ABS	off	on
				0		LFRR TC supply (VSC only) 0 for ABS	off	on

Command definitions and details are contained in the document / 命令的定义和细节在以下文件中:

20110524_Diagnostic_Proposal_for_CHANA_C201_Annex_1.9_To_CHANA(draft).doc

Example:

To create a pressure increase at FL for 70ms the command:

$31 $01$30 $28 $A8 $15 $00 $07 $AA $15

is activating

ABS iso valves RF,LR,RR	A8
TC iso LFRR, TC supply LFRR and Pump motor	15
waiting time = 70ms	00 07
all ABS iso valves	AA
TC iso LFRR, TC supply LFRR and Pump motor	15

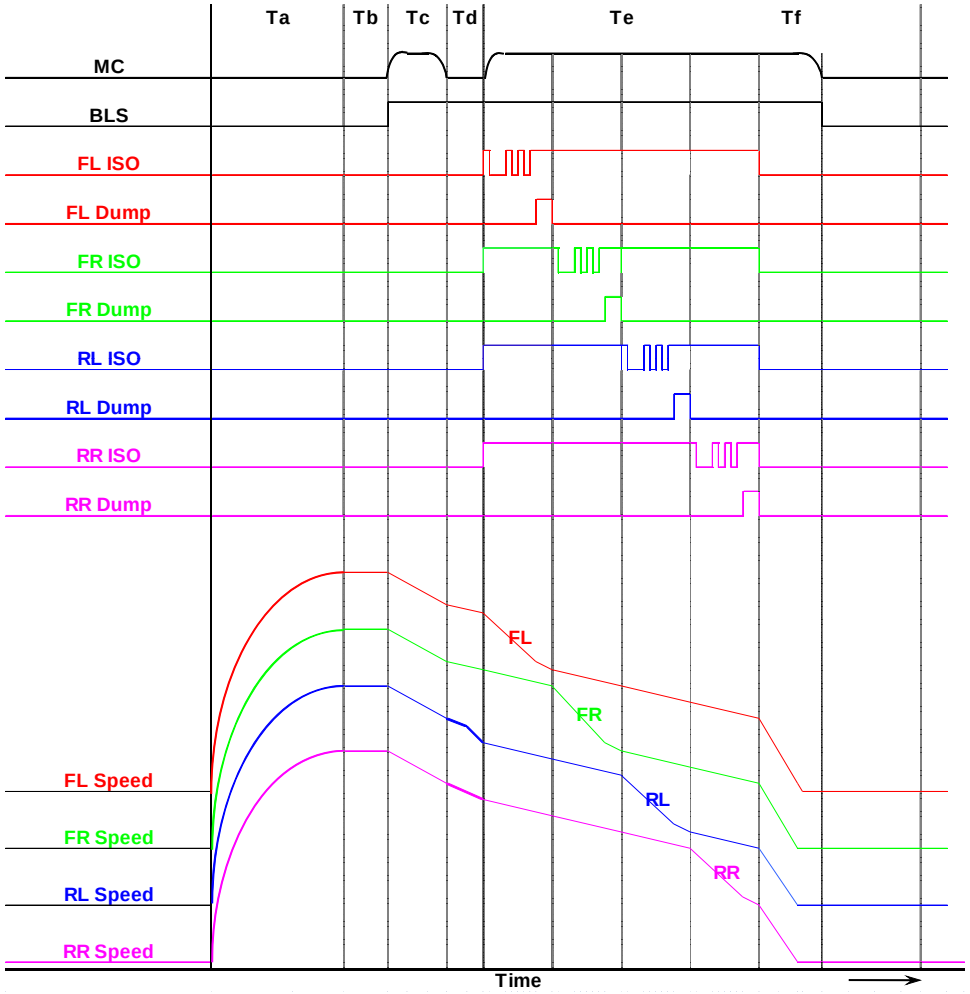
DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

EOL Procedure Proposal : Roller test bench 下线检测步骤建议:转鼓试验台

(High speed test / 高速测试)

System: TRW EBC460 ABS, with EPB



time table / 时间

		sec / 秒
Ta -	acceleration rollers coupled 连接加速转鼓设备	5
Tb -	decoupling rollers 120 kph 脱钩滚筒	2
Tc -	base brake test 基础制动和压力传感器测试	2
Td -	EPB Test 电子驻车制动测试	2
Te -	dynamic test 动态测试	6 - 10
Tf -	stop to zero 回到静止	

The duration time has to be evaluated at EOL Tester.
测试时间需要在下线检测设备上进行评估。

DRAFT

Note: The activation time of the valves has to be tested in the plant together with supplier of EOL machine.

注意：阀的激励时间需要在工厂同下线检测设备供应商一起来测试。

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

EOL Procedure Proposal: Roller test bench 下线检测步骤建议:转鼓试验台

System: TRW EBC460 ABS/ 系统：TRW EBC460 ABS

The operator keep the brake pedal **pressed** with constant force during the entire execution time.
操作人员需要在整车测试过程中以恒定的踏板力踩住制动踏板。

The tester software turns the valves of each wheel into pressure increase and pressure release cycles during the dynamic test.
诊断设备软件需要在动态测试过程中对各个车轮对应的阀进行压力上升和下降的控制。

The actual speed of each wheel are measured by the ECU and read by the tester, for further evaluation at the end of the test. (optional, only on request)
各个车轮的实际轮速通过ECU被测量及被诊断设备所读取，供测试结束时使用。（非必须项）

The timing of valve actuation and deceleration measuring times must be adapted onto the dynamics of the roller's test stand and the vehicle configuration.
电磁阀激励的时间和减速度测量次数必须按照下线检测设备及实际车辆配置进行调整。

The timing of the deceleration rates must be set according to the given measuring times and the initial braking applied.

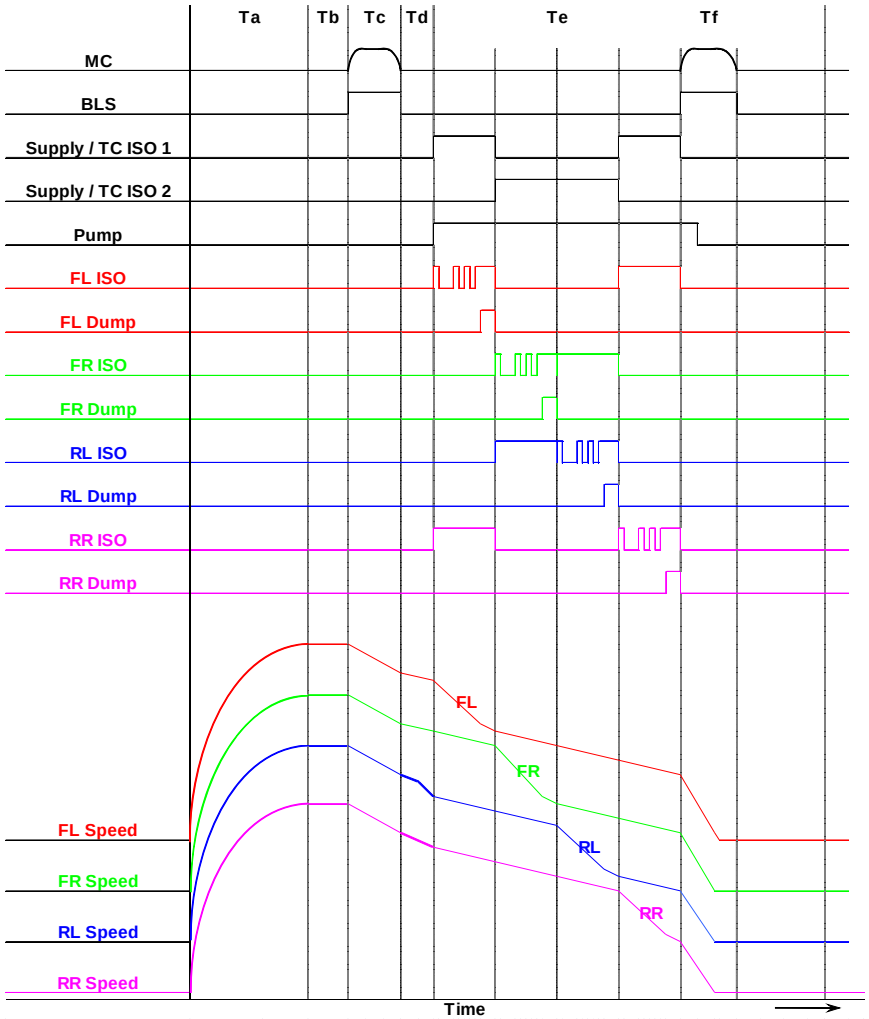
DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

EOL Procedure Proposal : Roller test bench 下线检测步骤建议:转鼓试验台

(High speed test / 高速测试)

System: TRW EBC460 ESC, with EPB



time table / 时间

		sec / 秒
Ta -	acceleration rollers coupled 连接加速转鼓设备	5
Tb -	decoupling rollers 120 kph 脱钩滚筒	2
Tc -	base brake test 基础制动和压力传感器测试	2
Td -	EPB Test 电子驻车制动测试	2
Te -	dynamic test 动态测试	6 - 10
Tf -	stop to zero 回到静止	

The duration time has to be evaluated at EOL Tester.
测试时间需要在下线检测设备上进行评估。

DRAFT

Note: The activation time of the valves has to be tested in the plant together with supplier of EOL machine.

注意：阀的激励时间需要在工厂同下线检测设备供应商一起来测试。

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011

EOL Procedure Proposal: Roller test bench 下线检测步骤建议:转鼓试验台

System: TRW EBC460 ESC / 系统：TRW EBC460 ESC

The operator keep the brake pedal **released** during the entire execution time.
操作人员需要在整车测试过程中**释放**制动踏板。

The tester software turns the valves of each wheel into pressure increase and pressure release cycles during the dynamic test.
诊断设备软件需要在动态测试过程中对各个车轮对应的阀进行压力上升和下降的控制。

The actual speed of each wheel are measured by the ECU and read by the tester, for further evaluation at the end of the test. (optional, only on request)
各个车轮的实际轮速通过ECU被测量及被诊断设备所读取，供测试结束时使用。（非必须项）

The timing of valve actuation and deceleration measuring times must be adapted onto the dynamics of the roller's test stand and the vehicle configuration.
电磁阀激励的时间和减速度测量次数必须按照下线检测设备及实际车辆配置进行调整

The timing of the deceleration rates must be set according to the given measuring times and the initial braking applied.

DRAFT

This copyrighted document is the property of TRW and is disclosed in confidence. It may not be copied, disclosed to others, or used for manufacturing, without the prior written consent of TRW.
© TRW Automotive Holdings Corp. 2011